

Question No:6

DS

Session

Queue

Question Information

Level 1 Challenge 56

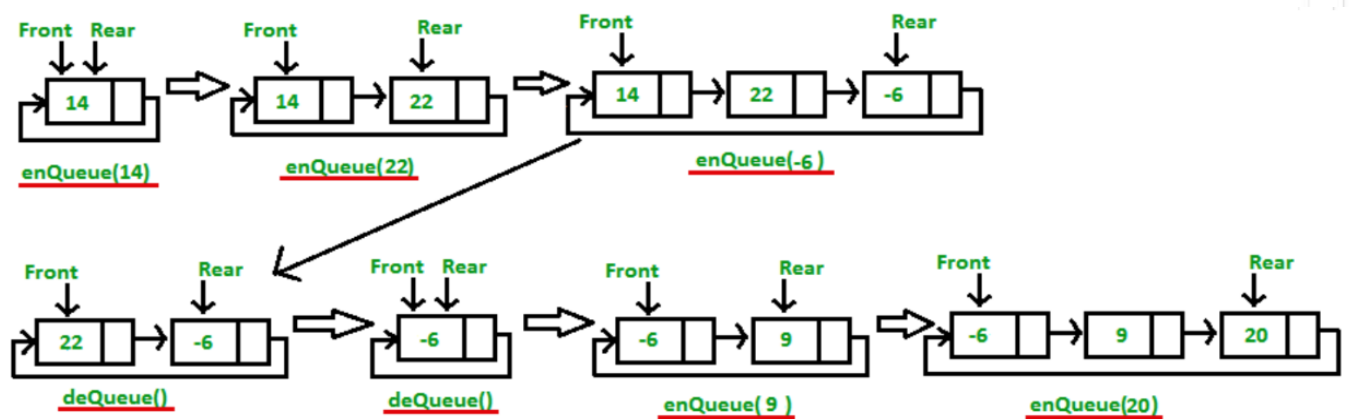
Question description

Lalitha is a B.Tech student. During her final year Campus Interview, she has an opportunity to get a job in a software company in Bangalore.

The company provides Five months training period with Rs.30000/month Package. Then it will be incremented to Rs.55000 per month.

At the end of the training, the examination was conducted for all freshers, Lalitha got a question paper and one of the questions comes under the concept of Queues in data structure that is Circular Linked List Implementation in Circular Queue.

Can you help?



Input Format

First line indicates the number of elements to be inserted in the queues.

Second line indicates the elements.

Output Format

first line indicates the dequeue element of Q2

Second line indicates the dequeue element of Q1

Third line line indicates the dequeue element of Q0

Note: Refer sample input and output test cases

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Logical Test Cases

Test Case 1

INPUT (STDIN)

```
10
1 2 3 4 5 6 7 8 9 10
```

EXPECTED OUTPUT

```
Elements in Circular Queue are:1 2 3 4 5 6 7 8 9 10
Deleted value = 1
Deleted value = 2Elements in Circular Queue are:3 4
5 6 7 8 9 10
```

Test Case 2

INPUT (STDIN)

```
10
16 17 18 19 20 11 12 13 14 15
```

EXPECTED OUTPUT

```
Elements in Circular Queue are:16 17 18 19 20 11 12
13 14 15
Deleted value = 16
Deleted value = 17Elements in Circular Queue are:18
19 20 11 12 13 14 15
```

Code:

```
1  #include <stdio.h>
2  #include <stdlib.h>
3  struct Node
4  {
5      int data;
6      struct Node *next;
7  };
8  struct Node *front = NULL;
9  struct Node *rear = NULL;
10 void enQueue(int data)
11 {
12     struct Node *newNode;
13
14     newNode = (struct Node *)malloc(sizeof(struct Node));
15
16     newNode->data = data;
17
18     if (front == NULL)
19     {
20         front = newNode;
21         rear = newNode;
22         newNode->next = front;
23     }
24     else
25     {
26         newNode->next = front;
27         rear->next = newNode;
28         rear = newNode;
29     }
```

```
30     }
31     int deQueue()
32     {
33         int value;
34         struct Node *temp;
35
36         if (front == NULL)
37         {
38             return -1;
39         }
40
41         /* Only one element */
42         if (front == rear)
43         {
44             value = front->data;
45
46             free(front);
47
48             front = NULL;
49             rear = NULL;
50
51             return value;
52         }
53
54         temp = front;
55         value = temp->data;
```

```
56
57     front = front->next;
58     rear->next = front;
59
60     free(temp);
61
62     return value;
63 }
64 void display()
65 {
66     struct Node *temp;
67
68     if (front == NULL)
69     {
70         return;
71     }
72
73     printf("Elements in Circular Queue are:");
74
75     temp = front;
76
77     do
78     {
79         printf("%d", temp->data);
80
81         temp = temp->next;
```

```

82
83         if (temp != front)
84         {
85             printf(" ");
86         }
87
88     } while (temp != front);
89
90     printf("\n");
91 }
92 int main()
93 {
94     int n;
95     int data;
96
97     scanf("%d", &n);
98
99     for (int i = 0; i < n; i++)
100    {
101        scanf("%d", &data);
102        enqueue(data);
103    }
104    /* Display original circular queue */
105    display();
106    /* Delete first element */
107    printf("Deleted value = %d\n", dequeue());
108    /* Delete second element */

```

```

109    printf("Deleted value = %d\n", dequeue());
110    /* Display remaining circular queue */
111    display();
112    return 0;
113 }

```

Output:

Test Case-1:

```

10
1 2 3 4 5 6 7 8 9 10
Elements in Circular Queue are:1 2 3 4 5 6 7 8 9 10
Deleted value = 1
Deleted value = 2
Elements in Circular Queue are:3 4 5 6 7 8 9 10

```

Test Case-2:

```
10
16 17 18 19 20 11 12 13 14 15
Elements in Circular Queue are:16 17 18 19 20 11 12 13 14 15
Deleted value = 16
Deleted value = 17
Elements in Circular Queue are:18 19 20 11 12 13 14 15
```

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